



Review article

Mortality and morbidity risks associated with floods: A systematic review and meta-analysis[☆]Zhengyu Yang, Wenzhong Huang, Joanne E. McKenzie, Pei Yu, Ke Ju, Yao Wu, Bo Wen, Yuming Guo, Shanshan Li^{*}

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ABSTRACT

Background: Flood disasters are projected to increase in severity, duration, and frequency in the context of climate change, and the risks of mortality and morbidity may increase after floods, which will overwhelm health systems.

Objective: This study aims to synthesize current epidemiological evidence about the impacts of floods on mortality and morbidity.

Method: After performing a systematic literature search from 2000 to 2023, we included studies involving human participants, with exposures of floods, and with outcomes of mortality or morbidity.

Results: In total, 37 studies were included in evidence syntheses. Meta-analyses yielded an overall relative risk of 1.26 (95% confidence interval [CI]: 1.10, 1.46), 1.10 (1.08, 1.13), 1.11 (1.04, 1.20), and 1.38 (1.18, 1.62) for all-cause mortality and morbidities of overall gastrointestinal diseases, diarrhea diseases, and dysentery, respectively. Although meta-analyses were not conducted, evidence from at least three studies consistently supported that exposure to floods was associated with increased risks of malaria and respiratory diseases. The evidence for other outcomes was reported but either limited or uncertain.

Conclusion: This study suggests that exposure to floods is associated with increased risks of all-cause mortality and morbidities of overall gastrointestinal diseases, diarrhea diseases, dysentery, malaria, and respiratory diseases, while further research is urgently called.

1. Introduction

As the most frequent natural disaster worldwide, floods have dramatic and extensive adverse impacts on the environment, economy, and human health (Hu et al., 2018a). The number of people affected by floods has been increasing in the past four decades (Hu et al., 2018a). As estimated, there are 1.81 billion people (23% of the world's population) directly exposed to 1-in-100-year flood disasters (Rentschler et al., 2022). In the context of climate change, flood disasters are projected to increase in severity, duration, and frequency due to more frequent extreme precipitation events and rising sea levels (Tellman et al., 2021).

Commonly, official records of health burden arising from floods have only considered deaths caused by the direct physical forces (e.g., drowning) or hazards (e.g., electrocution and hypothermia) of floods (Yan et al., 2020; Lindersson et al., 2023). However, these assessments

have not considered the substantial number of deaths and diseases resulting from non-accidental causes. For example, non-accidental deaths and diseases may result from contamination of food and drinking water, exposure to pathogens (i.e., fungi, bacteria, and viruses), impaired access to health services, and displacement (Alderman et al., 2012; Dosa et al., 2020; Paterson et al., 2018). This suggests that the health impacts of floods are under-studied. Comprehensively understanding the impact of floods on mortality and morbidity is important for improving policies, including disaster response strategies and health system adaptation.

Of the studies that have examined the relationship between flood exposure with mortality and morbidity, the key gaps identified include the following (Zhong et al., 2018). First, most of the original studies focused on mental disorders, and the findings for other outcomes were mixed (Zhong et al., 2018). To our knowledge, no systematic review nor

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meta-analysis was conducted for the association of flood exposure with mortality and morbidity other than mental disorders. There were two mapping reviews on this association, which primarily described the research pattern and landscape in the field through systematic literature searches but without findings synthesis and study quality assessment (Alderman et al., 2012; Zhong et al., 2018). In addition, one of them only included studies published during 2004–2011 (Alderman et al., 2012), while the other one did not consider short-term health impacts (i.e., <6 months), but included studies published over a more recent and broader time period (i.e., before 2016) (Zhong et al., 2018). Therefore, we aimed to undertake a systematic review with meta-analyses to address the gaps. Specifically, to examine the impacts of floods on mortality and morbidity.

2. Methods

This systematic review was reported in accordance with the PRISMA statement and was registered on PROSPERO (CRD4202339981). (Page et al., 2021).

2.1. Eligibility criteria

For this systematic review, studies were eligible if they were studies including humans, with exposures included floods, and with outcomes included mortality or morbidity. We excluded studies that only included life quality, external cause of deaths (e.g., injury and suicide), symptoms (e.g., cough, runny nose, rash, sore throat), or psychological health (e.g., post-traumatic stress disorder) as outcomes; and studies focusing on disasters that may lead to floods (e.g., hurricanes and earthquakes). We excluded studies based on the type of report including reviews, meta-analyses, descriptive studies, response letters, and conference abstracts. Studies in all languages were considered for inclusion. We only included eligible studies published from 2000 January 1st to 2023 March 10th. When studies with the same outcomes, same flood events, and similar time frames had overlapping populations, only the study with the largest population was included.

2.2. Search strategy

OVID Medline, EMBASE, and Scopus were searched to identify eligible studies. We did not filter the searches by language. The search strategy consisted of search keyword searches and corresponding subject headings (if available) for flood and mortality or morbidity. The search strategies are available in Section 1 of Appendix. The references cited in relevant reviews identified in the literature search and publications linked to conference abstracts were also examined. Authors were contacted for additional information.

2.3. Screening and selection

Two researchers (ZY and WH) independently screened the titles and abstracts of records identified through the literature search using the PICO Portal (<https://www.picoportal.org/>). Only records that were excluded by both researchers were excluded at this stage. Full-text of potentially eligible studies were retrieved, and screened independently by two researchers (ZY and WH). Disagreements on eligibility were resolved by consulting a senior researcher (PY).

2.4. Data extraction

Data extraction was initially piloted by three researchers (ZY, WH, and PY) on a sample of four studies to ensure consistency of data extraction and to modify the data extraction where required. Following piloting, two researchers (ZY and WH) independently performed the data extraction. From each study, we extracted author names, publication year, outcomes, outcome ascertainment method, exposure

assessment method, control group, source of the study population, flood events, study type, time frame of impacts (e.g., infectious morbidity was assumed to change during 0–14 days post exposure to flood), statistical analysis methods (e.g., adjustment for lag and covariates), effect estimates (and their confidence intervals). In studies where effects were reported for multiple time frames (e.g., 0–7 days and 0–14 days post flood exposure), the estimate of the longest time frame (i.e., 0–14 days) was extracted. Discrepancies and uncertainty were resolved in regular meetings of the three researchers (ZY, WH, and PY).

2.5. Assessment of risk of bias

Two researchers (ZY and WH) independently assessed the risk of bias (RoB) of each study using the National Institutes of Environmental Health Sciences National Toxicology Program Office of Health Assessment and Translation (OHAT) tool (Rooney et al., 2014). As recommended by developers of the OHAT tool, we tailored all criteria for our topic (see Table S1). Using these criteria, key domains (i.e., exposure assessment, outcome assessment, and confounding bias) and other domains (selection bias, attrition/exclusion bias, and other sources of bias) were judged as being at a definitely low, probably low, probably high, or definitely high RoB. Studies were judged to be at a “definitely high RoB” when they had at least one key domain with definitely high RoB and at least four domains with a “probably high” or “definitely high” RoB. Studies judged to have a “definitely high” RoB were excluded from any synthesis (qualitative or quantitative). Discrepancies and uncertainty were resolved in regular meetings of the three researchers (ZY, WH, and PY).

2.6. Data syntheses

To determine whether a conclusion can be reached regarding the association of flood exposure with mortality and morbidity (Russo, 2007), inverse variance-weighted random-effects meta-analysis was used to combine relative risk (RR) estimates when three or more RRs with comparable time frames were available for a particular outcome (Wilker et al., 2023). We combined RR estimates based on two time frames, namely, <6 months and ≥6 months (Zhong et al., 2018). Estimates representing a single day (e.g. 11th day post exposure to the flood) or not measured by RRs were not included in the meta-analysis, but are reported. Odds ratios approximate RRs for rare outcomes (i.e. with event rates <10%) and so were treated as RRs and included in the meta-analyses in this circumstance (Sedgwick, 2014; Grimes and Schulz, 2008). We chose the random-effects model because we expected there would be diversity in population, setting and study design across studies (Borenstein et al., 2010). We used the restricted maximum likelihood estimator for between study variability and the Knapp-Hartung method for confidence intervals (Wilker et al., 2023; Borenstein et al., 2010). We quantified inconsistency using the I^2 statistic and formally tested for heterogeneity using Cochran's Q test. We used forest plots to visually display the effect estimates from studies included in meta-analyses and for those not included in a meta-analysis (i.e., when there fewer than three estimates for a particular outcome and timeframe). We undertook post-hoc sensitivity analyses to examine the impact on the meta-analysis estimates of excluding studies with at least three domains with probably/definitely high RoB and by using fixed-effect models.

3. Results

3.1. Studies characteristics

Of the initially identified 5871 records, 1914 were duplicates, and 3859 were excluded at the title/abstract screening stage (Fig. 1). From the 98 full-text articles retrieved, 50 studies met the inclusion criteria (Table S2); however, 13 were judged to be at a definitely high RoB, and were excluded from the qualitative and quantitative syntheses

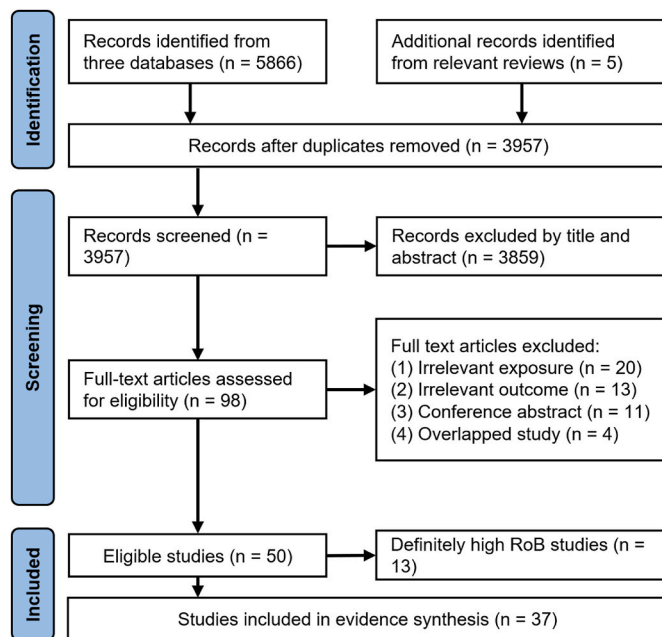


Fig. 1. PRISMA flow diagram of identifying eligible studies. Abbreviation: RoB = risk of bias.

(Table S2). Consequently, a total of 37 studies were synthesized (Fig. 1). These studies represented 17 countries/territories in eight sub-regions. Most (26/37) studies were conducted in Asia. More than half (20/37) of the studies only covered flood events within one year (Fig. 2B). Almost all studies were restricted to the impacts within one state/province (Table 1), except for five estimating the impacts of flood events across Ireland (Boudou et al., 2021), mainland China (Guo et al., 2020), Thailand (Suwanpakdee et al., 2015), Vietnam (Phung et al., 2021), and the United Kingdom (Milojevic et al., 2011), respectively. Time-series study was the most widely used study design (Fig. 2C).

3.1.1. Outcome ascertainment

A variety of morbidities were assessed as well as all-cause, cardiovascular (CVD), and respiratory mortality. Except for two studies ascertaining outcomes with standardized tools (Rodríguez-Llanes et al., 2011, 2016), and three through self-report (Mulder et al., 2019; Milojevic et al., 2012; Pradhan et al., 2007), almost all (32/37) studies were based on administrative datasets (Table 1). Among the studies employing an administrative dataset, some (9/32) only covered a part of the general population because the datasets were from a simple source (e.g., datasets of hospital visits, emergency visits, or admissions); however, most (23/32) were based on a comprehensive source (e.g., diseases surveillance network, whole-population insurance database, or mortality registration) and so comprehensively captured the incidences of outcomes in the general population.

3.1.2. Exposure assessments

The exposure assessment method varied importantly across studies (Table 1). Almost all studies defined the exposed population as those residing in flooded areas during flooded periods. On the temporal scale, exposure to floods was defined as days, weeks, months, seasons, years, or pregnancy periods, with the flood events based on disaster records (Table 1). Using disaster records or interviews, exposure to floods was defined on a spatial scale as individuals residing in households (Milojevic et al., 2012; Miyamori et al., 2023; Yoshida et al., 2021; Vollaard et al., 2004), communities/village/district/town (Boudou et al., 2021; Suwanpakdee et al., 2015; Milojevic et al., 2011; Rodríguez-Llanes et al., 2011, 2016; Pradhan et al., 2007; Chen et al., 2021a; Elsanousi et al., 2018; Vanasse et al., 2016; Deng et al., 2022), cities

(Yan et al., 2020; Hetherington et al., 2021; Liu et al., 2016, 2019, 2020; Zhang et al., 2016a, 2016b, 2019; Hu et al., 2018b; Huang et al., 2016; Wade et al., 2014; Ni et al., 2014; Lan et al., 2022), or provinces with flood events (Guo et al., 2020; Phung et al., 2021; Ma et al., 2021; Liu et al., 2017; Gao et al., 2016; Tong et al., 2011). The data used for these assessments were from sources including official records, insurance certifications, and surveys/interviews. Almost all studies in mainland China determined flooded areas and periods by referring to the records provided by Yearbooks of Meteorological Disasters, where flood periods were defined as days with cumulative rainfall of >80 mm for three consecutive days (Ma et al., 2021). Studies in Japan defined flooded households as those destroyed or whose primary family caregiver was seriously injured or lost jobs due to flood events, which were certified by a government-run insurance system (Miyamori et al., 2023; Yoshida et al., 2021). A study in Canada ascertained flooded areas using water level data from governmental hydraulic records (Vanasse et al., 2016). Flooded households were identified through interviews in three studies (Mulder et al., 2019; Milojevic et al., 2012; Vollaard et al., 2004). Some studies used a distinctive definition of flood exposure. A study in Uganda defined flooded villages as those containing or were along flooding rivers (Boyce et al., 2016); a study in the USA defined exposure group as the population residing in postcode areas where at least one-third of the area was flooded (Quist et al., 2022); a study in Cambodia quantified exposure as square kilometers of floods in the district of residence (Saulnier et al., 2018).

3.1.3. Risk of bias

We judged exposure assessment and confounding as the main risks of bias (Table S2), with 13 and 18 included studies judged as at a probably or definitely high RoB for these domains respectively. Potential bias in the exposure assessments arose from the time frames used (weeks/months/seasons/years), which could have resulted in misclassifying the periods before flood events as exposed periods. Potential bias from confounding arose from inadequate control of confounders in the analysis.

3.2. Associations with mortality

Three studies investigated the risk of all-cause mortality (i.e., deaths from all causes) related to floods in the <6 month time interval (Fig. 3). (Yan et al., 2020; Milojevic et al., 2012; Pradhan et al., 2007) The meta-analysis of the three studies yielded an overall RR of 1.26 (1.10, 1.46), where the time frames of the estimates varied from four days to 24 weeks, with some inconsistency ($I^2 = 30\%$, p -value of test for heterogeneity = 0.24). Only one study in the UK examined the effect of flood in the ≥ 6 month time interval, finding that the risk of all-cause mortality decreased in one year (RR 0.90 [0.82, 0.99]) (Milojevic et al., 2011). For specific mortality outcomes (Fig. 4), the risk of CVD mortality was found to increase during 0–4 days post floods (Yan et al., 2020). However, no significant association was found for respiratory mortality (Yan et al., 2020).

3.3. Associations with morbidity

The non-external hospitalization risk was found increased (RR 1.07 [1.03, 1.10]) during the four weeks post floods in Vietnam (Phung et al., 2021). Another study focused on the impacts of a flood event in Japan (2018) on population aged >45 years. The authors observed that the risk of nursing home admission was three times higher among those affected by floods compared with those not affected by floods (Fig. 4). (Miyamori et al., 2023) This finding suggested that older people might be more vulnerable to the health impacts of floods.

3.3.1. Gastrointestinal diseases

Sixteen studies estimated the impacts of floods on the risks of gastrointestinal diseases. In addition to overall gastrointestinal diseases,

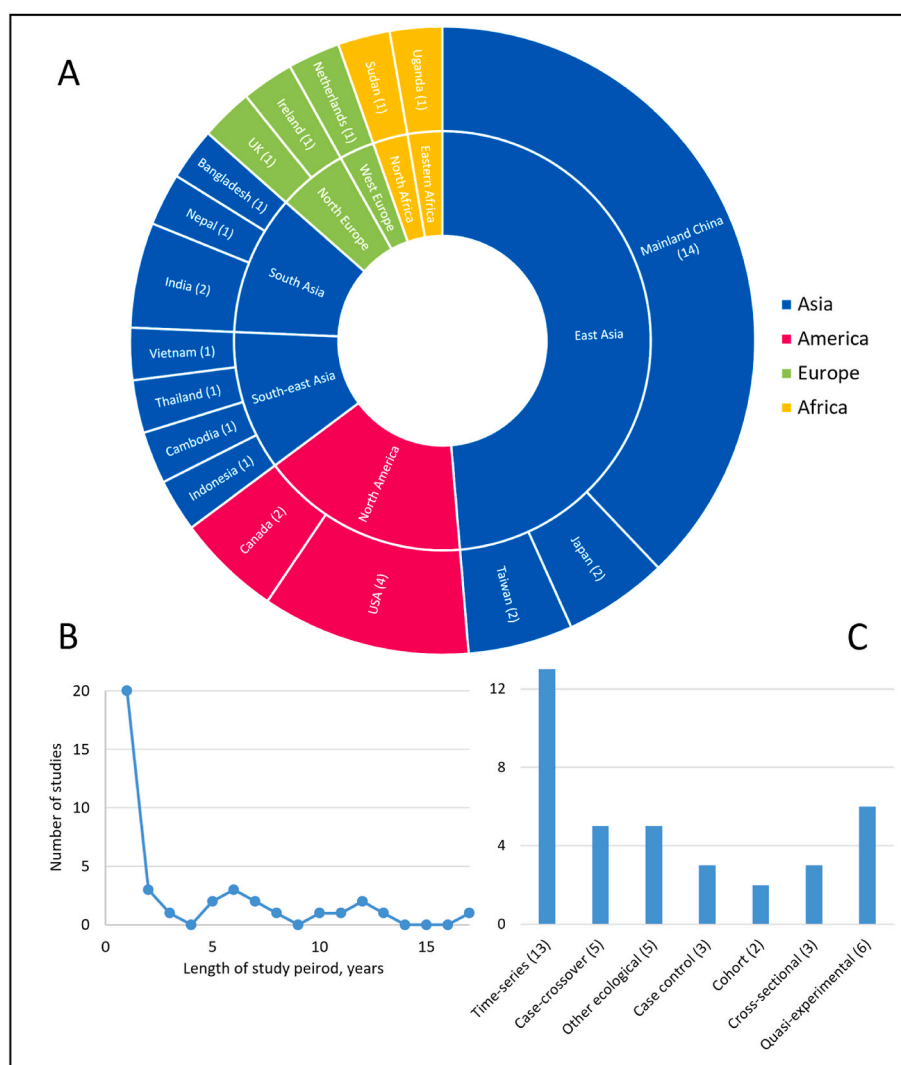


Fig. 2. Numbers of studies by country/territory and sub-region according to the United Nations geoscheme (A), by length of study period (B), and by study design (C). Cross-sectional studies were not included in the summary of length of study period.

the studies examined dysentery, diarrhea, acute gastroenteritis, verotoxigenic *E. coli* (VTEC) enteritis, and cryptosporidiosis (Figs. 3–4).

Three studies estimated the risk of overall gastrointestinal diseases in the <6 month time interval (specifically within three weeks) following two flood events in 2016 and 2018 in the USA (Quist et al., 2022), 129 events during 2002–2008 in the USA (Wade et al., 2014), and 240 events during 1998–2008 in Taiwan (Huang et al., 2016), respectively. The meta-analysis yielded an overall RR of 1.10 (1.08, 1.13), with consistent RRs ($I^2 = 0\%$, p -value of test for heterogeneity = 0.55).

Four studies estimated the impact on diarrhea diseases incidence in the <6 month time interval (Milojevic et al., 2012; Liu et al., 2019; Lan et al., 2022). The meta-analysis of the effect estimates yielded an overall RR of 1.11 (1.04, 1.20), with no evidence of between-study heterogeneity ($I^2 = 0\%$, p -value of test for heterogeneity = 0.47). Among the studies not included in the meta-analysis, the study in Cambodia reported that the risk of diarrhea increased in the current month and third month post flood periods (Saulnier et al., 2018), and the study in China showed that the risk increased on 11th days post the flood event in 2007 (Gao et al., 2016). Only a study in China estimated the effect of flood in the ≥ 6 month time interval and found that the risk of diarrhea increased after a flood event in 2016 (Zhang et al., 2019).

Seven studies estimated the effects on dysentery in the <6 month time interval (Liu et al., 2016, 2017, 2019; Zhang et al., 2016b; Ni et al.,

2014; Ma et al., 2021; Gao et al., 2016). Nine effect estimates were extracted because two study provided effect estimates for two locations and two flood events, respectively (Liu et al., 2016, 2017). The meta-analysis yielded an overall RR of 1.38 (1.18, 1.62). Large between-study heterogeneity was observed ($I^2 = 87\%$, p -value < 0.01); however, all the RRs indicated flood events were associated with an increased risk of dysentery.

Only one study investigated the impact on acute gastroenteritis risk (Mulder et al., 2019). This study found an increased risk of acute gastroenteritis (RR 4.20 [2.10, 8.40]) two weeks post a flood event in the Netherlands. On the other hand, the risks of enteritis caused by VTEC (RR 1.35 [1.25, 1.46]) and cryptosporidiosis (RR 1.57 [1.27, 1.48]) were found to increase a year post a flood event in 2015 summer in Ireland (Boudou et al., 2021).

3.3.2. Infectious diseases

Ten studies (Fig. 4) investigated the impacts on the risks of infectious diseases. Studies found the hospitalization risk of infectious diseases increased (RR 1.15 [1.03, 1.29]) during 0–4 weeks post floods in Vietnam (Phung et al., 2021), and the hospitalization risk of food/water-borne infectious diseases increased during 0–14 days post floods (Deng et al., 2022).

The impacts on the risks of typhoid and paratyphoid in <6 month

Table 1
Characteristics of included studies.

	Study	Flood events and locations	Study design	Study population	Case number and sample size	Exposure Assessment	Control group	Outcomes	Outcome assessment
1	Miyamori et al. (2023) (Miyamori et al., 2023)	Flood from 2018 to 06 to 2018–07 in 3 prefectures in Japan	Cohort	Almost >65 years old (98%)	6423 cases from 187861 participants	Care insurance users whose homes were destroyed or primary family caregiver seriously injured or lost job	Individuals in non-impacted household	Nursing home admissions	Administrative dataset: simple source
2	Quist et al. (2022) (Quist et al., 2022)	Floods caused by Hurricanes Matthew and Florence in 2016 and 2018 in North Carolina	Quasi-experimental	General population	868691 cases from all local population	Exposed area: a ZIP code with at least one third area flooded based on flood extent data from the NC Department of Public Safety; exposure period: 3 weeks from the day of hurricane landfall	ZIP code with less than one third area flooded	Acute gastrointestinal	Administrative dataset: comprehensive source
3	Lan et al. (2022) (Lan et al., 2022)	Floods in 2017–2019 in Sichuan, China	Time-series	General population	124602 cases from all local population	The ten-day periods with flooding events were defined as exposed periods according to the Yearbooks of Meteorological Disasters in China	Days without floods	Diarrheal	Administrative dataset: comprehensive source
4	Deng et al. (2022) (Deng et al., 2022)	Floods in 2002–2018 in New York State	Time-series	General population	NA cases from all residents	Days, locations, and types of floods were obtained from the Storm Events Database maintained by the NOAA	Days without floods	CVD, chronic respiratory diseases (asthma and COPD), respiratory infection, food/water-borne diseases hospitalization	Administrative dataset: simple source
5	Yoshida et al. (2021) (Yoshida et al., 2021)	Flood from 2018 to 06 to 2018–07 in 3 prefectures, Japan	Cohort	Almost >65 years old (98%)	23654 cases from 264614 participants	Care insurance users whose homes were destroyed or primary family caregiver seriously injured or lost job	Individuals in non-impacted household	Cognitive decline incidence	Administrative dataset: simple source
6	Phung et al. (2021) (Phung et al., 2021)	Flood season in 2011 in seven provinces in Vietnam	Time-series	General population	753029 non-external hospitalizations	Flooded days were defined as those within the flood season. All provinces were regarded with the same exposure	Days without flood season	Non-external, infectious, and respiratory hospitalization	Administrative dataset: simple source
7	Ma et al. (2021) (Ma et al., 2021)	All floods in 2005–2016 in Chongqing, China	Time-series	General population	45691 cases	Days with floods, ascertained by the Yearbooks of Meteorological Disasters in China. Precipitation data were used when the administrative regions of a flood event was not available.	Days without floods	Bacillary dysentery	Administrative dataset: comprehensive source

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Table 1 (continued)

	Study	Flood events and locations	Study design	Study population	Case number and sample size	Exposure Assessment	Control group	Outcomes	Outcome assessment
8	Hetherington et al. (2021) (Hetherington et al., 2021)	Flood in 2013 Calgary, Canada	Quasi-experimental	Pregnant women & new births	1455 preterm birth, 2696 SGA, 854 preeclampsia, and 1098 gestational hypertension from 18291 births	Flooded period was the flooded date, and flooded areas are all communities of three highly impacted cities	Regions without floods, periods without floods	Preterm birth, SGA, gestational hypertension, and preeclampsia	Administrative dataset: comprehensive source
9	Chen et al. (2021) (Chen et al., 2021a)	All floods in 2003–2015 across Taiwan	Case-crossover	Children/adolescent	55488 ER visits from the whole population in Taiwan	Flooded weeks and townships were determined using information acquired from National Science and Technology Center for Disaster Reduction	Weeks without floods	Atopic dermatitis emergency room visit	Administrative dataset: comprehensive source
10	Boudou et al. (2021) (Boudou et al., 2021)	2015/16 winter flood across Ireland	Other ecological	General population	83 VTEC cases, 238 cryptosporidiosis cases from all local population	Cases were exposed if the centroids of their residential Small Area were covered by flood extent. Flood period was defined as July 1, 2015 to July 1, 2016	Areas without floods	VTEC enteritis and cryptosporidiosis	Administrative dataset: comprehensive source
11	Yan et al. (2020) (Yan et al., 2020)	Flood on 2012-07-21 in Beijing, China	Other ecological	General population	NA cases from all local population	Exposure period was 2012 July 21.	Days without floods	Mortality, CVD, and respiratory mortality inpatient and outpatient admission	Administrative dataset: comprehensive source
12	Liu et al. (2020) (Liu et al., 2020)	Floods in 2005–2012 in Mengshan Guangxi China	Time-series	General population	1070 cases from all local population	The ten-day periods with flooding events were defined as exposed periods according to the Yearbooks of Meteorological Disasters in China	10-day periods without floods	Acute haemorrhagic conjunctivitis	Administrative dataset: comprehensive source
13	Guo et al. (2020) (Guo et al., 2020)	1998 Yangtze River flood across 31 provinces in China	Quasi-experimental	Children/adolescent	1131 cases from 108175 births	Four exposure groups were determined based on birth dates and the flood period (i. e., the worst flooded 3 months): post-partum exposed, prenatal exposed, preconception exposed, and unexposed cohorts. Participants' residential provinces were classified as not flooded, mild flooding, moderate flooding, or severe flooding	Individuals in provinces without floods and in the unexposed cohort	Cognitive impairment	Self-reported
14	Zhang et al. (2019) (Zhang et al., 2019)	Flood in 2016 in Anhui, China	Quasi-experimental	General population	274621 cases in flooded areas, 129356 cases in	Post-intervention period: 1 year	Pre-intervention period: 3	Infectious diarrhea	Administrative dataset:

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Table 1 (continued)

	Study	Flood events and locations	Study design	Study population	Case number and sample size	Exposure Assessment	Control group	Outcomes	Outcome assessment
					non-flooded areas from local population	after the flood event; during-flood period: 2016-06-18 to 2016-08-31	years before the flood event		comprehensive source
15	Mulder et al. (2019) (Mulder et al., 2019)	2015 summer flood in 60 locations in Netherlands	Cross-sectional	General population	87 AGE and 166 ARI from 1158 participants	Self-reported contact types	Individuals in non-flooded households	Acute gastroenteritis and acute respiratory infection	Administrative dataset: simple source
16	Liu et al. (2019) (Liu et al., 2019)	Six floods in 2006–2010 in Guangxi, China	Time-series	General population	45131 cases from local population	Flooded days of each city were determined by the Yearbooks of Meteorological Disasters in China	Days without floods	Diarrhea, dysentery, typhoid, and paratyphoid	Administrative dataset: comprehensive source
17	Saulnier et al. (2018) (Saulnier et al., 2018)	Floods in 2008–2013 in 11 districts in Cambodia	Time-series	General population	NA cases from 204186 inpatients, 5544051 outpatients	Square kilometers of flood water in each district in each month were estimated using data from the MODIS sensors	NA	Diarrhea, acute respiratory infections, and skin infections	Administrative dataset: comprehensive source
18	Hu et al. (2018) (Hu et al., 2018b)	Five floods in 2009–2013 in Qingdao, China	Time-series	Children/adolescent	55920 cases from local population	Months with floods were determined using the Yearbooks of Meteorological Disasters in China	Months without floods	Hand-foot-mouth disease	Administrative dataset: comprehensive source
19	Elsanousi et al. (2018) (Elsanousi et al., 2018)	Flood in August 2013 in Gezira, Sudan	Other ecological	General population	5069, 5549, and 7262 cases from 990247, 1019953, and 1049659 in 2011, 2012, and 2013	The 3 months since the flood event were exposure period	The same months in the two preceding years	Malaria	Administrative dataset: comprehensive source
20	Liu et al. (2017) (Liu et al., 2017)	Floods in 2004–2010 in Guangxi, China	Time-series	General population	78794 cases from local population	Months with floods were determined using the Yearbooks of Meteorological Disasters in China	Months without floods	Bacillary dysentery	Administrative dataset: comprehensive source
21	Zhang et al. (2016a) (Zhang et al., 2016a)	Floods in 2007–2012 in Nanchong, Sichuan, China	Case-crossover	General population	370 cases from local population	Days with floods were determined using the Yearbooks of Meteorological Disasters in China	Days without floods	Japanese encephalitis	Standardized tool
22	Zhang et al. (2016b) (Zhang et al., 2016b)	Flood in 2007 in Zibo Shandong, China	Case-crossover	General population	902 cases from local population	Days with floods were determined using the Yearbooks of Meteorological Disasters in China	Days without floods	Bacillary dysentery	Administrative dataset: comprehensive source
23	Vanasse et al. (2016) (Vanasse et al., 2016)	Flood in spring 2011 in Saint-Jean-sur-Richelieu, Canada	Case-control	Adults	1170 cases from 78582 participants	One flooded area was defined using water level data from the hydraulic records of the Government of Canada. The period of the flood event is	Two areas with similar population sizes, comparable SESs, and presence of a river not affected by the flood. The same period	Acute cardiovascular diseases	Administrative dataset: comprehensive source

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Table 1 (continued)

	Study	Flood events and locations	Study design	Study population	Case number and sample size	Exposure Assessment	Control group	Outcomes	Outcome assessment
						the exposure period	in year before and after were control period		
24	Rodriguez-Llanes et al. (2016) (Rodriguez-Llanes et al., 2016)	Floods in 2006 and 2008 in four blocks of Odisha, India	Cross-sectional	Children <5 years-old	871 participants from 34 villages	Exposure to floods was defined as living in a village flooded by both flood events based on records from the Odisha State Disaster Mitigation Authority	Individuals in villages not impacted by both flood events	Wasting, stunting, underweight	Administrative dataset: comprehensive source
25	Liu et al. (2016) (Liu et al., 2016)	Flood in 2012 in Jishou and Huaihua in Hunan, China	Case-crossover	General population	184 cases from local population	Days with flood were defined using cumulative rainfall data: cumulative rainfall of more than 80 mm for three consecutive days or 250 mm for ten consecutive days	Days without floods	Bacillary dysentery	Administrative dataset: comprehensive source
26	Huang et al. (2016) (Huang et al., 2016)	Floods in 1998–2008 in Taiwan	Time-series	General population	NA cases from whole population in Taiwan	Days with floods for each county/city were determined using flood records obtained from the Central Weather Bureau	Days without floods	Diseases of eyes, skins, and gastrointestinal diseases	Administrative dataset: comprehensive source
27	Gao et al. (2016) (Gao et al., 2016)	Flood in 2007 at Huai River in Anhui, China	Time-series	General population	5904 malaria cases, 2681 diarrhea cases, 1148 dysentery cases, 111 Japanese encephalitis cases, 104 HAV cases, 55 HEV cases, 16 typhoid and paratyphoid cases, and 4 HFRS cases from local population.	Days with floods were determined using the Yearbooks of Meteorological Disasters in China	Days and areas without floods	Malaria, diarrhea, dysentery, japanese encephalitis, HAV, HEV, typhoid and paratyphoid, and HFRS	Administrative dataset: comprehensive source
28	Boyce et al. (2016) (Boyce et al., 2016)	Flood in May 2013 in Bugoye subcounty in the Western Ugandan Highland	Quasi-experimental	General population	2315 cases from 7596 participants	Post-flood period: 12-month after the month of flood event. Flooded villages: the place where a flood-affected river within or along the boundaries	Pre-flood period: 12-month before the month of flood event; unflooded village	Malaria	Administrative dataset: simple source
29	Suwanpakdet al. (2015) (Suwanpakdee et al., 2015)	Floods during 2010–2012 across Thailand	Time-series	General population	4940 cases from local population	Months with floods were determined for each districts using the spatial information that were provided by the Geo-Informatics and Space Technology	Months without floods	Leptospirosis	Administrative dataset: comprehensive source

(continued on next page)

Table 1 (continued)

	Study	Flood events and locations	Study design	Study population	Case number and sample size	Exposure Assessment	Control group	Outcomes	Outcome assessment
30	Wade et al. (2014) (Wade et al., 2014)	Floods in 2002–2008 in Massachusetts, USA	Case-crossover	General population	270457 admissions from local population	Development Agency Days with floods were determined using the information from the Storm Events Database for each county/town	Days without floods	Gastrointestinal diseases emergency room visit	Administrative dataset: comprehensive source
31	Ni et al. (2014) (Ni et al., 2014)	Floods in 2004–2009 in Henan, China	Time-series	General population	24536 cases from local population	Flooded, non-flooded, pre-flooded, and post-flooded months were determined using the Yearbooks of Meteorological Disasters in China	Non-flooded months	Dysentery	Administrative dataset: comprehensive source
32	Milojevic et al. (2012) (Milojevic et al., 2012)	Flood in 2004 in Matlab, Bangladesh	Quasi-experimental	General population (mortality and diarrhea), children (acute respiratory infection)	5280 deaths, 4250 diarrhea cases, and 23163 acute respiratory infection cases from 211139 participants	Exposed individuals: the subjects were exposed if the floor of any household in the bari had been under water during the flood period. Flooded period: pre-flood, during-flood, and post-flood periods were defined	Unexposed individuals and pre-flood period	All-cause mortality, diarrhea, and acute respiratory infection	Administrative dataset: simple source
33	Tong et al. (2011) (Tong et al., 2011)	Red River flood in 1997 in North Dakota, USA	Case-control	Pregnant women & new births	2477 low birth weight, 4753 preterm birth, and 3573 SGA from 55344 births	The flooded period was defined by referring the disaster declaration: pre-disaster is non-exposed, post-disaster is exposed	Pre-disaster period	Low birth weight, preterm birth, and small for gestational age	Administrative dataset: comprehensive source
34	Rodriguez-Llanes et al. (2011) (Rodriguez-Llanes et al., 2011)	Floods in 2006 and 2008 in Orissa district, India	Cross-sectional	Children <5 years-old	111 stunting, 61 underweight, and 42 wasting from 352 participants	Exposed villages: flooded by both flood events in 2006 and 2008	Villages not flooded by both events	Malnutrition	Administrative dataset: comprehensive source
35	Milojevic et al. (2011) (Milojevic et al., 2011)	Floods in 1994–2005 in England and Wales	Other ecological	General population	1464 deaths from local population	Flooded postcode areas were determined using the historic flood event outlines from the UK Environment Agency's National Flood and Coastal Defence Database. Years with flood were defined for each postcode area. If a flood event happened in the same postcode within 2 years since last flood event, this flood	Flooded region, pre-flood year.	All-cause mortality	Administrative dataset: simple source

(continued on next page)

Table 1 (continued)

	Study	Flood events and locations	Study design	Study population	Case number and sample size	Exposure Assessment	Control group	Outcomes	Outcome assessment
36	Pradhan et al. (2007) (Pradhan et al., 2007)	Flood around July 19, 1993 in Nepal	Other ecological	General population	302 deaths from 41501 participants	event would be removed Rest of the year since flood events were defined as the exposed period	The preceding year	All-cause mortality	Administrative dataset: comprehensive source
37	Vollaard et al. (2004) (Vollaard et al., 2004)	Floods in 2001–06 to 2003–02 in Jakarta, Indonesia	Case-control	General population	93 cases, 289 fever control, and 378 community control	Self-reported: individuals were exposed if their home was inundated in the 12 months preceding the interview	Individuals in non-inundated households	Typhoid fever and paratyphoid fever	Administrative dataset: comprehensive source

Notes: administrative datasets included those collected from simple source (e.g., datasets of hospital visits, emergency visits, or admissions) or from comprehensive source (e.g., diseases surveillance network, whole-population insurance database, or mortality registration). *Studies were regarded as with definitely high RoB through RoB assessment. Abbreviations: NA = not available; USA = the United States of America; UK = the United Kingdom; CVD = cardiovascular diseases; VTEC = verotoxin-producing *E. coli*; ER = emergency room; SGA = small for gestational age; ARI = acute respiratory infection; HAV = hepatitis A; HEV = hepatitis E; HFR = hemorrhagic fever with renal syndrome; NA = not available.

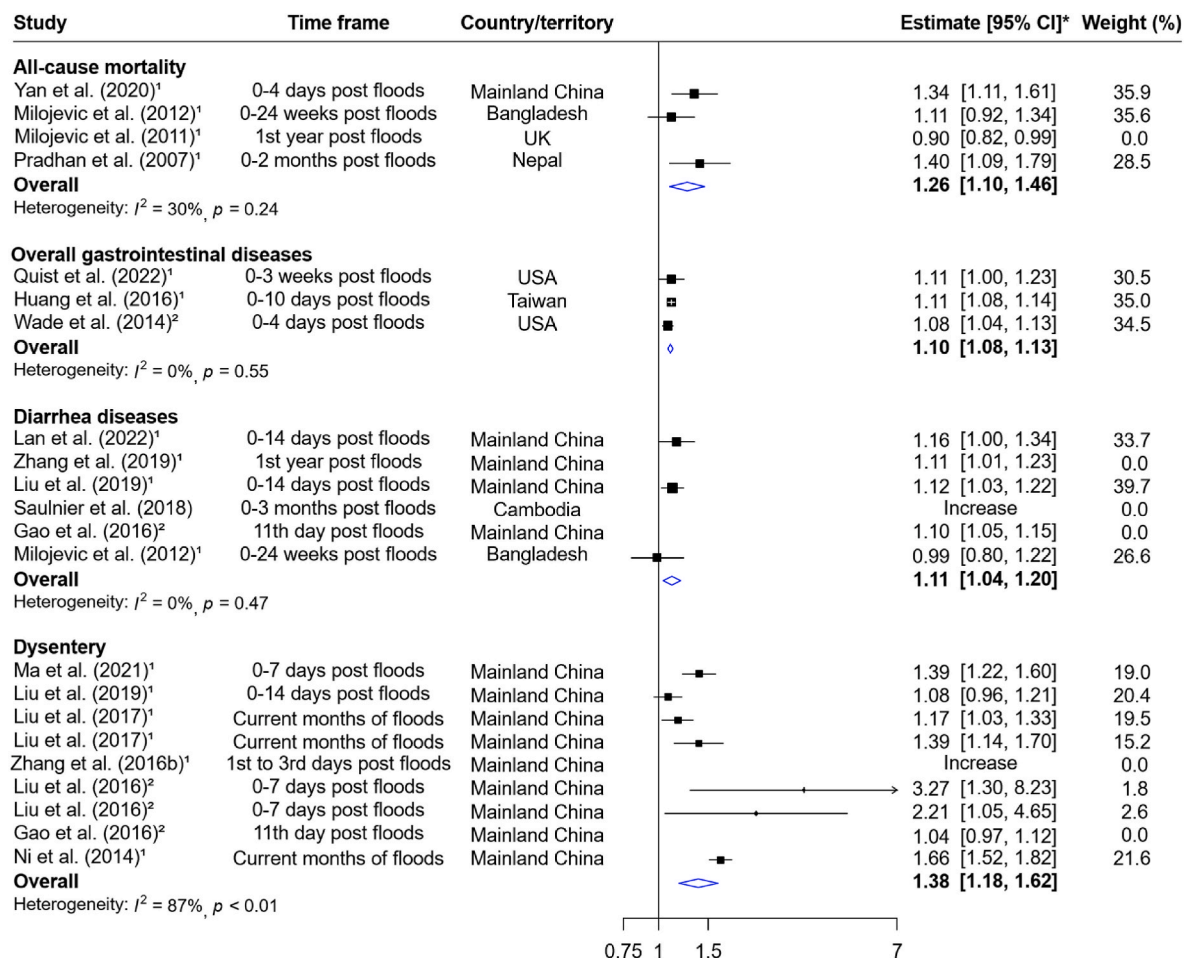


Fig. 3. Forest plots for meta-analyses of relative risks of all-cause mortality, diarrhea, dysentery, and gastrointestinal diseases associated with flood exposure. Weights represent the contribution of each study to the overall meta-estimate. Abbreviation: UK = the United Kingdom; USA = the United States of America; CI = confidence interval. The estimate is ¹relative risk or ²odds ratio. *Estimates from Milojevic et al. (2011) and Zhang et al. (2019) were not combined with other estimates because of their incomparable time-frames (i.e., time frame >6 months); estimates from Gao et al. (2016), Zhang et al. (2016b), and Saulnier et al. (2018) were not combined with other estimates because of its incomparable effect measurement; two effect estimates for different flood events from Liu et al. (2017) and two effect estimates for different locations from Liu et al. (2016) were included.

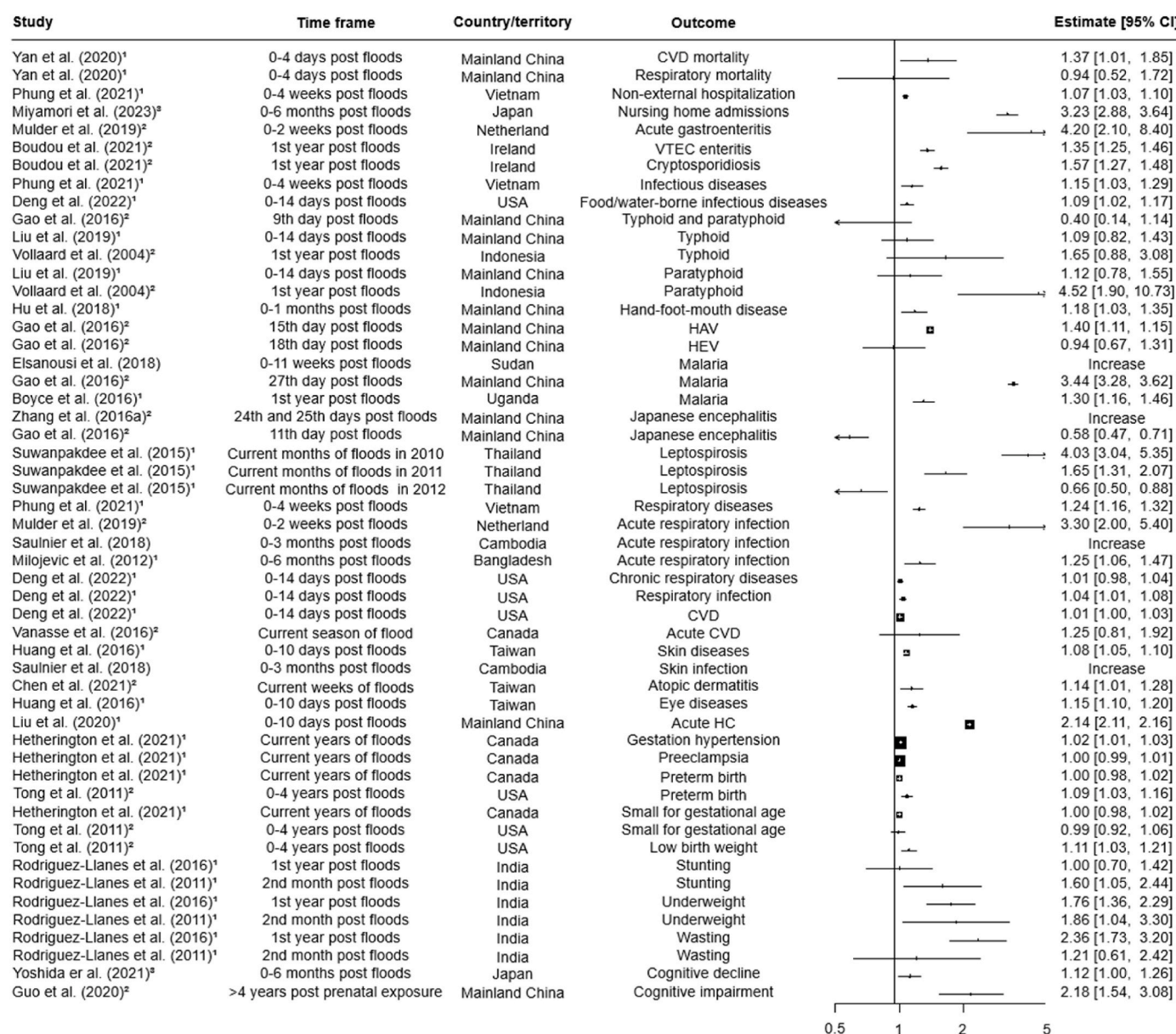


Fig. 4. Forest plots and estimates of the associations of flood exposure with mortalities and morbidities for each study and outcome. Abbreviations: UK = the United Kingdom; USA = the United States of America; CVD = cardiovascular diseases; VTEC = verotoxigenic *E. coli*; HAV = hepatitis A; HEV = hepatitis E; HC = hemorrhagic conjunctivitis; CI = confidence interval. The estimate is ¹relative risk, ²odds ratio, or ³hazard ratio. *The effect estimates from [Elsanousi et al. \(2018\)](#) and [Saulnier et al. \(2018\)](#) were not measured by relative risks. [Suwanpakdee et al. \(2015\)](#) reported three effect estimates for the same outcome (leptospirosis) but for different flood events.

time interval were investigated by two studies, which were both conducted in mainland China ([Liu et al., 2019](#); [Gao et al., 2016](#)). With all of the confidence intervals included the null value, no significant association was found. The association between floods and the risk of typhoid in one year in a city in Indonesia was not statistically significant (RR 1.65 [0.88, 3.08]); however, there was an increased risk of paratyphoid (RR 4.52 [1.90, 10.73]) ([Vollaard et al., 2004](#)).

Hand-foot-mouth disease was investigated in one study in mainland China ([Hu et al., 2018b](#)), which found an increased risk during 0–1 month post exposure to floods (RR 1.18 [1.03, 1.35]). While the risk of HAV increased (RR 1.40 [1.11, 1.15]) on the 15th day post a flood event in Anhui, mainland China, no significant association was found between this flood event and HEV (RR 0.94 [0.67, 1.31]) ([Gao et al., 2016](#)).

The risk of malaria incidence was increased after flood events in Sudan, mainland China, and Uganda, although the time frames varied. In Sudan, the incidence risk of malaria during 0–11 week post a flood

event was higher than the risk during the same period in the 2 years preceding the event ([Elsanousi et al., 2018](#)); in mainland China, the risk reached a peak on the 27th day post a flood event (RR 3.44 [3.28, 3.62])⁴⁸; and the risk was higher (RR 1.30 [1.16, 1.46]) in flooded villages, compared to non-flooded villages in the year following a flood event in Uganda ([Boyce et al., 2016](#)).

The impact on Japanese encephalitis was investigated in two studies in mainland China. The authors reported that the smallest and largest risks of Japanese encephalitis occurred on 11th and 24th days post floods in Anhui (RR 0.58 [0.47, 0.71]) ([Gao et al., 2016](#)) and Sichuan (2.00 [1.14, 3.52]) ([Zhang et al., 2016a](#)), respectively.

Only one study estimated the association of floods with human leptospirosis incidence, covering flood events during 2010–2012 in Thailand ([Suwanpakdee et al., 2015](#)). The associations across flooding events were found to be inconsistent across the years: 2010 (RR 4.03 [3.04, 5.35]), 2011 (RR 1.65 [1.31, 2.07]), and 2012 (RR 0.66 [0.50,

0.88]).

3.3.3. Respiratory diseases

Five studies provided evidence for respiratory diseases from Vietnam, Netherlands, Cambodia, Bangladesh, and the USA, consistently suggesting the risk of respiratory diseases would increase after floods, although the outcomes varied among all respiratory diseases, acute and chronic respiratory diseases, and respiratory infection (Fig. 4). (Phung et al., 2021; Mulder et al., 2019; Milojevic et al., 2012; Deng et al., 2022; Saulnier et al., 2018)

3.3.4. Cardiovascular diseases

One study in the USA indicated that the hospitalization risk of CVD increased (RR 1.01 [1.00, 1.03]) during 0–14 days post floods (Deng et al., 2022). However, no significant association was found between flood and the risk of acute CVD (1.25 [0.81, 1.92]) (Vanasse et al., 2016).

3.3.5. Skin and eye diseases

Three studies explored the associations between flood and skin diseases (Fig. 4). (Chen et al., 2021a; Huang et al., 2016; Saulnier et al., 2018) One study estimated the risk of overall skin diseases (Huang et al., 2016), while the other studies estimated the risks of skin infection and atopic dermatitis. All studies found an increased risk of skin diseases post flood events. The time frames varied from a week to three months post floods.

The associations for eye diseases were estimated in two studies. One study in Taiwan found an increased risk of overall eye diseases during 0–10 days post flood exposure (Huang et al., 2016). Another study in mainland China found a twofold increased risk of acute hemorrhagic conjunctivitis in 0–10 days post floods (Liu et al., 2020).

3.3.6. Adverse birth and pregnancy outcomes

The associations of floods with adverse birth outcomes were estimated in two studies (Fig. 4). One study in Canada estimated the associations in the year of a flood event (Hetherington et al., 2021), and the study in the USA estimated the association during 0–4 years post a flood event (Tong et al., 2011). The associations for preterm birth and small for gestational age were estimated by both studies. In the two studies, the risk of preterm birth was found to increase (RR 1.09 [1.03, 1.16]) during 0–4 years post the flood event in the USA, while the other associations were statistically insignificant. Additionally, the risk of low birth weight was also found to increase (RR 1.11 [1.03, 1.21]) during 0–4 years post the flood event in the USA.

3.3.7. Malnutrition

The risk of malnutrition among children aged 6–59 months was estimated by Rodriguez-Llanes et al. in two studies in a district in Orissa, India (Fig. 4). (Rodriguez-Llanes et al., 2011, 2016) These studies were based on the same population and flood events, but with different time frames and sample sizes. They found that the risks of stunting and being underweight were higher in exposed group in the second month post exposure; however the association for wasting was statistically insignificant (Rodriguez-Llanes et al., 2011). At one year after the flood event, they found that there was an increased risk of being underweight and wasting (Rodriguez-Llanes et al., 2016).

3.3.8. Cognitive dysfunction

Two studies investigated the impacts of floods on cognitive dysfunction (Fig. 4). One estimated the impact on the risk of cognitive decline among elderly population (i.e., >65 year-old) during 0–6 months post a flood event in Japan (Yoshida et al., 2021), and found an increased risk among the exposed group (RR 1.12 [1.00, 1.26]). The other study assessed the impact of the 1998 Yangtze River Flood on the risk of cognitive impairment among 108175 four to eight year-old children across 31 provinces in mainland China (Guo et al., 2020).

This study found an increase risk among children that were exposed to the flood event during perinatal period.

4. Discussion

This is the first systematic review for the associations of flood with mortality and morbidity. We synthesized findings from previous studies after study quality assessment. We found that flood exposure was associated with increased risks of all-cause mortality and morbidities of overall gastrointestinal diseases, diarrhea diseases, dysentery, malaria, and respiratory disease after flood exposure. Evidence for other health outcomes was limited or uncertain.

4.1. Health impact pathways

Floods can result in diseases and deaths in several potential pathways. First, floods can contaminate water supply systems (Barbetta et al., 2022), through which the risks of gastrointestinal diseases can increase, and the transmission of some infectious diseases (e.g., typhoid and paratyphoid) can be facilitated (Lee et al., 2020). Second, floods can create breeding grounds for fungi, bacteria, viruses, and vectors (e.g., mice, cockroaches, and mosquitoes) (Suhr and Steinert, 2022). When this occurs, outbreaks of infectious diseases, including plague, malaria, Japanese encephalitis, and leptospirosis may be triggered (Suhr and Steinert, 2022). Third, flood events can force people to evacuate, leading to displacement. Even though temporary accommodation may be provided, hygiene issues still commonly occur from poor access to sanitation facilities (Paterson et al., 2018). Hence, the risks of water-borne diseases, infectious diseases, and infections can rise. Displacements also mean limited access to medical and health facilities, which may lead to delayed diagnoses and treatments, more complications, exacerbated prognosis, and consequently elevated mortality; lacking protection from environmental hazards (e.g., non-optimal temperatures), which impact a broad range of health conditions (Zhao et al., 2021; Chen et al., 2021b; Wu et al., 2022); and high densities of fungi, bacteria, virus, and vectors in limited rooms (Lee et al., 2020). Fourth, mortality and morbidity can be exacerbated by disaster-related psychological disorders via many mechanisms including weakening immune systems, sleep problems, substance abuses, and poor self-care (Trief et al., 2006; Sharpe and Davison, 2021).

4.2. Research gaps and future perspectives

4.2.1. Potential effect modifications

Considering the health impact pathways of floods, the health effects of flood may be modified by location- (e.g., socioeconomic status, dietary habit, sewage system, disaster response strategies, medicine sources), climate- (e.g., climate zone, temperature, and humidity), biology- (e.g., density of mosquitos, vegetation types), and flood-related (e.g., severity, duration, and flood types) factors, as well as vulnerabilities of populations (e.g., children, elderly population, and people with pre-existing chronic diseases). Understanding these effect modifications helps identify the most vulnerable populations, which is essential for targeting protections and preventions. Furthermore, the understanding can facilitate predicting the health effects in places where health data are not available, mostly low-income countries, which could be used to estimate flood-attributable health burden to optimize resource allocations.

4.2.2. Temporal and spatial coverages

We found that the temporal (i.e., flood events) and spatial coverages (i.e., locations) of almost studies were limited. This means that the generalizations of these effect estimates were compromised. To obtain more reliable estimates, studies with a broad temporal and spatial coverages are warranted.

4.2.3. Time frame and lag effects

Most studies only arbitrarily/empirically assumed a time frame for the lag effect, without examining the time frames statistically. Lag effects for gastrointestinal diseases and infectious diseases have been statistically examined by some studies, considering the breeding, incubation, and latent periods of pathogens (Liu et al., 2019; Zhang et al., 2016a; Wang et al., 2021). The findings consistently suggested that the risk of gastrointestinal diseases reached the peak on several days post flood exposure (Liu et al., 2016, 2017, 2019; Zhang et al., 2016b; Ni et al., 2014; Ma et al., 2021). The impacts on mortality and morbidity may be delayed because the pathways between flood and human health are not immediate. However, the lag effects were understudied for most health impacts. Identifying the lag-response associations related to flood exposure is a key to coping with elevated mortality and morbidity risks after flood events.

4.3. Clinical and policy implications

Both health service providers and policymakers should be aware of the increased health risks post floods, when health/medicine service demands will sharply rise. Considering that the capability and accessibility of health systems are generally disrupted during floods, the unfulfillment for the increasing health/medicine service demands will be exacerbated, especially in communities poor in health/medicine resources or with infrastructures vulnerable to floods (Romanello et al., 2023). To optimize prognosis, reduce avoidable deaths from natural causes, and ease flood-attributable health burden, health service providers should strengthen capacities in medical supplies, human resource management, and triage strategy; and policymakers should periodize enhancing health system resilience to natural disasters, as well as improving disaster preparedness, early warning and detection systems, and efficient disaster response protocols to reduce the hazardous exposure during and post floods.

4.4. Conclusions

Previous studies had examined an extensive range of mortality and morbidity risks related to floods, although studies were scarce in almost all countries/territories. We undertook a systematic review examining the associations between floods and health outcomes and found associations of flood exposure with elevated risks of all-cause mortality and morbidities of overall gastrointestinal diseases, diarrhea diseases, dysentery, malaria, and respiratory diseases. These findings suggest that medical practitioners and policymakers should raise awareness and preparation for the elevated burden on health systems after flood events. The evidence for the morbidities of other infectious diseases, cardiovascular diseases, skin and eye diseases, adverse birth and pregnancy outcomes, malnutrition, and cognitive dysfunction was either limited or uncertain. Future studies examining more flood events and locations, effect modifiers, and the lag effects of flood exposure with mortality and morbidity are needed.

CRediT authorship contribution statement

Zhengyu Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Wenzhong Huang:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Joanne E. McKenzie:** Writing – review & editing, Supervision, Methodology, Investigation. **Pei Yu:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Ke Ju:** Writing – review & editing, Investigation. **Yao Wu:** Writing – review & editing, Investigation. **Bo Wen:** Writing – review & editing, Data curation. **Yuming Guo:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Shanshan Li:** Writing – review & editing, Supervision,

Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Ethical approval

Ethical approval was not needed because this study does not involve human participants.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors did not use any generative AI or AI-assisted technology.

Consent to publish

Consent was not needed because this study does not involve human participants.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2024.120263>.

Data availability

Template data collection forms, data extracted from included studies, data used for all analyses, analytic code, and any other materials used in the review were available upon request.

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